The Fault Is in the Substrate: Substrate Identity as the Dominant Driver of Coupling in Bounded Predictive Systems

Abstract

Any bounded predictive system, by virtue of compressing the world it models, must encode a prior; such a prior is a bias; and systems sharing a substrate must therefore share constraints on their viable priors. We test the empirical signature of this claim using two large language models (Mistral and Llama-3.3-70B, accessed via Mistral and Groq APIs respectively) as substrates, each instantiating five personas with classically distinct astrological framings on 120 verified natal charts from the Astrodatabank corpus. Within-substrate persona pairs exhibit mean pairwise mutual information of 0.103 nats; cross-substrate pairs exhibit 0.041 nats. The cross-substrate value is approximately 60% lower, with non-overlapping bootstrap 95% confidence intervals. The persona-matched test (same prompt, different substrate) yields cross-substrate mutual information substantially below within-substrate mutual information across different prompts on four of five personas: substrate identity beats prompt identity as a driver of output coupling. The effective number of independent observers in the combined ten-observer ensemble is 2.08—approximately the number of distinct substrates, not of nominal personas. We argue that this result, together with a partial information decomposition showing that substrate-shared observers share what they predict more than how uncertain they are, constitutes operational evidence for a Peircean structural realism: "objective reality" for any community of bounded observers is the asymptote of substrate-coupled convergence under disciplined methodology, not a substrate-independent quantity such convergence approximates. We extend the framework to human cognition as a falsifiable prediction and discuss what the position commits us to and what it does not.

Keywords: bounded cognition; predictive processing; partial information decomposition; substrate-coupling; structural realism; large language models; ensemble methodology

1 Introduction

In 1952, Carl Jung and the statistician Karl Liljequist undertook what Jung called the "astrological experiment." They collected birth charts of 483 married couples and tested whether classical synastry aspects—Sun–Moon conjunctions between partners in particular—occurred more frequently in real marriages than in randomly paired control samples drawn from the same population [Jung, 1952]. The first batches produced striking results, with probabilities for some configurations reaching one in ten million. As the experiment progressed and the researchers' enthusiasm waned, the effect attenuated. By the time all 483 marriages had been analyzed, the statistical signal had collapsed to chance. Jung's interpretation, framed in his theory of synchronicity, was that the effect was real but engagement-dependent: a phenomenon in which observer participation was constitutive rather than incidental to the result.

This pattern has recurred across every subsequent large-scale test of natal astrology. Carlson [1985], in a double-blind *Nature* study, found astrologers performed at chance on personality–chart matching. Dean et al. [2016]'s multi-decade review of hundreds of studies reaches the same conclusion. Across many independent tests, on different chart features against different outcomes, the marginal signal from chart to outcome converges to zero as sample size grows. We accept this finding without qualification. Our pre-registered experiment, reported below, replicates it: top-1 accuracy on a six-class profession prediction task using natal charts lies at chance across all observers tested.

The contribution of this paper is to ask a different question. Granting that the marginal signal is null, what is happening in the structured *between-observer agreement* that the marginal null does not measure? Jung's interpretation invoked synchronicity. We propose a different reading: that what changed across Jung's experiment was not the world but the observers' priors, and with them the projection of the world into the observers' outputs. The structured component that remained, even at the marginal null, is on the framework developed here the operational signature of substrate-shared compression in bounded cognitive systems. It is measurable. It is testable. And, we will argue, it is what disciplined inquiry into "objective reality" can be said to converge upon—not as an approximation to a substrate-independent truth, but as the asymptote of substrate-coupled agreement under methodologically disciplined conditions.

The argument develops as follows. By the rate-distortion theorem [Shannon, 1948], any bounded predictive system must compress the world it models; any compression entails a prior; any prior is a bias. Bias is therefore not a contaminant of an observer's measurement function but constitutive of it. From this it follows that systems sharing a substrate—a neural architecture, an evolutionary history, a training corpus—must share structured constraints on their viable priors, and these shared constraints must manifest as a coupling term between the systems' outputs that is measurable as mutual information.

We test this prediction directly. Five personas with classically distinct astrological framings (Hellenistic, Psychological, Vedic, Evolutionary, Uranian) are run on two substrates: Mistral (open-mistral-nemo) and Llama-3.3-70B (accessed via the Groq API). Each persona scores the same 120 charts from the same six profession categories. We compute the full 10×10 matrix of pairwise mutual information between observer outputs, decompose ensemble information geometry using partial information decomposition, and quantify the effective number of independent observers via a coupled-oscillator order parameter.

The empirical finding is straightforward. Cross-substrate observer pairs decouple sharply from within-substrate pairs (mean MI of 0.041 vs. 0.103 nats, non-overlapping bootstrap CIs). The persona-matched test—same prompt, different substrate—shows that running an identical Hellenistic-tradition prompt across Mistral and Llama produces *less* coupling than running the Hellenistic and Vedic prompts on the same Mistral instance. Substrate identity beats prompt identity as a driver of statistical agreement on every persona but one. The effective number of independent observers in the combined ten-observer ensemble is 2.08—approximately the number of distinct substrates, not of nominal personas.

The philosophical move we make from this result is the following. If bounded cognition is necessarily bias-conditioned, and if shared substrate produces measurable and sharp coupling between substrate-sharing observers' outputs, then the question of an observer-independent measurement does not arise as a problem of contamination but as a structural impossibility for any bounded predictive system. We commit, accordingly, to the position that what we call objective reality is the asymptote of substrate-coupled convergence among observers sharing substrate, under disciplined methodology designed to maximize that convergence—not a substrate-independent quantity such convergence approximates. We see no obvious reason this should fail to apply to human cognitive systems and welcome arguments for why it would not.

The remainder of the paper proceeds as follows. Section 2 situates the framework against rate-distortion theory, the free-energy principle, and existing observer-dependence literatures in quantum measurement and philosophy of science. Section 3 describes the experimental design. Section 4 reports the cross-substrate result, the persona-matched test, and the information decomposition. Section 5 develops the philosophical implications and the extension to human cognition. Section 6 addresses limitations honestly. Section 7 concludes.

2 Bounded Cognition and the Necessity of Priors

The argument requires three pieces. First, the formal necessity of priors in any bounded predictive system. Second, the reading of bias as constitutive rather than contaminant. Third, the bridge from individual observer bias to inter-observer coupling under shared substrate.

2.1 Compression entails a prior

The rate-distortion theorem [Shannon, 1948] establishes that any finite encoder reproducing a source under a fidelity constraint must induce an equivalence class on inputs—it must treat distinguishable inputs as equivalent at some level of representation. The choice of equivalence class is the prior. There is no neutral encoder. A bounded predictive system—one with finite representational capacity—cannot represent the world without selecting what to retain and what to discard, and that selection cannot be made except under a prior over what matters.

The free-energy principle [Friston, 2010] extends this to biological systems: organisms minimize the divergence between predicted and observed sensory states, and the generative model that supports prediction encodes priors over the world. Clark [2013] develops the broader predictive-processing frame in which cognition is fundamentally prediction-under-prior. None of these literatures treats the prior as eliminable. The question is not whether a bounded system has priors but which priors it has, and how they relate to the priors of other systems.

2.2 Bias as constitutive, not contaminant

Standard treatments of observer effects—experimenter expectancy [Rosenthal, 1966], theory-ladenness of observation [Hanson, 1958, Kuhn, 1962]—frame the observer's biases as a problem to be acknowledged, modeled, or controlled. The implicit picture is that there exists, in principle, an observer-independent measurement function from which the contaminated observer-dependent measurement departs.

The framework developed here inverts this picture. Given the formal necessity of priors for any bounded encoder, there is no observer-independent measurement function to depart from. The observer's outputs are bias-conditioned all the way down. The question of an observer-independent measurement does not arise as a problem of contamination but as a structural impossibility for bounded systems. This position is adjacent to relational quantum mechanics [Rovelli, 1996], which establishes observer-relative state assignments in quantum measurement, but extends the relational framing from quantum to macroscopic bounded predictive systems.

2.3 Shared substrate as a coupling term

The novel claim is the following. If observers are bias-conditioned predictive systems, and if their biases are determined by their substrate—the architecture and history that fixes which priors are viable—then observers sharing substrate must share structured constraints on their viable priors. These shared constraints will produce statistically coupled outputs even when the observers are operating under nominally different surface configurations (different prompts, different framings, different interpretive traditions).

The coupling is measurable. For any two observers O_i, O_j producing outputs on a common stimulus set, the pairwise mutual information

$$I(O_i; O_j) = \sum_{a,b} \hat{p}(a, b) \log \frac{\hat{p}(a, b)}{\hat{p}(a)\hat{p}(b)}$$

quantifies the structured agreement between their outputs. When this quantity exceeds what would be observed under independent matched-marginal nulls, the excess is attributable to non-trivial shared structure. If our framework is correct, the excess should be substantially larger for observer pairs sharing substrate than for pairs not sharing substrate.

The empirical machinery for testing this is well-developed. Ameri et al. [2015] and Wood et al. [2020] establish mutual information as an order parameter for synchronization in coupled-oscillator systems, supporting the use of normalized coupling ratios to derive an effective number of independent observers N_{eff} from a nominally K-observer ensemble. Williams & Beer [2010] and Bertschinger et al. [2014] develop partial information decomposition (PID) machinery that further decomposes ensemble mutual information into redundant, unique, and synergistic components, allowing fine-grained characterization of *how* substrate-sharing observers share information.

3 Methods

3.1 Dataset

We assembled 120 natal birth charts as a stratified subsample from the Astrodatabank corpus maintained by Astrodienst, restricted to entries with Rodden rating AA or A (verified birth time or quoted from biography). The subsample contains 20 charts per profession category across six categories: scientist, artist, athlete, politician, writer, business. Profession categorization follows Astrodatabank's own taxonomy. Natal planetary positions and house placements were computed using the Swiss Ephemeris (Placidus houses, with Whole Sign placements for cross-check). The subsample was drawn deterministically from a 600-chart parent dataset with seed 42.

Natal astrology is used here not as a target of validation but as a deliberately ambiguous symbolic system over which trained observers can plausibly disagree. The chart-to-profession mapping itself, in our and prior analyses [Dean et al., 2016, Carlson, 1985], exhibits null marginal signal, which is precisely what makes it a clean substrate for isolating observer coupling from observer accuracy.

3.2 Substrates and personas

Two substrates were used:

- **Mistral** (open-mistral-nemo), accessed via the Mistral API.
- **Llama-3.3-70B**, accessed via the Groq API.

These were selected as the most divergent free-tier substrates available: distinct architectures, distinct training corpora, distinct organizational and engineering lineages. Within each substrate, five personas were instantiated via distinct system prompts encoding classically distinct interpretive traditions in natal astrology:

- O_1 (Hellenistic) — classical Greek tradition: rulerships, sect, traditional dignities, seven classical planets.
- O_2 (Psychological) — modern Jungian/archetypal tradition: planets as inner figures, outer-planet aspects weighted heavily.
- O_3 (Vedic) — jyotisha tradition: emphasis on Moon, ascendant, Jupiter and Saturn, bhava lords.
- O_4 (Evolutionary) — Jeffrey Wolf Green tradition: Pluto and lunar nodes central, profession as soul vehicle.
- O_5 (Uranian) — Hamburg School tradition: hard outer-planet aspects, tight orbs, midpoint structures.

The full text of all five system prompts is provided in supplementary materials. The five prompts are byte-identical across the two substrates. Each persona was asked, for each chart, to score the six professions on a 0–100 scale according to its interpretive framework. Total predictions collected: $2 \times 5 \times 120 = 1200$. Decoding parameters (temperature = 0.7, max output tokens = 200) and prompt format were held constant across substrates. Charts were presented in fixed order across personas to control for context-window effects; persona prompts were issued in independent API sessions to prevent cross-persona contamination.

3.3 Analyses

For each ordered pair of observers (O_i, O_j) in the combined 10-observer ensemble we computed:

Pairwise mutual information. The empirical $I(O_i; O_j)$ in nats, using top-1 profession predictions across the 6×6 outcome space.

Permutation null. For each pair, 1000 permutations of each observer's predictions independently across charts, recomputing MI under the matched-marginals null of independence. Z-scores were computed per pair.

Bootstrap confidence intervals. For aggregate statistics (within-substrate mean MI, cross-substrate mean MI, their difference), bootstrap resampling over the 120 charts (1000 iterations) yielded 95% confidence intervals.

Persona-matched MI. For each of the five personas p, we report $I(\text{Mistral}_p; \text{Llama}_p)$ (cross-substrate same persona) and compare it to the mean $I(\text{Mistral}_p; \text{Mistral}_{q \neq p})$ and $I(\text{Llama}_p; \text{Llama}_{q \neq p})$ (within-substrate different persona).

Total correlation scaling. For each subset size $k \in \{2, 3, 4, 5\}$ within each substrate, mean TC across subsets; for the combined 10-observer ensemble, the full joint TC.

Effective number of observers. Following the convention from coupled-oscillator literature [Ameri et al., 2015], $N_{\text{eff}} = K/[1 + (K-1)c]$ where $c = \text{TC}/(K \cdot \bar{H})$ is the normalized coupling ratio and $\bar{H}$ is the mean single-observer entropy.

Partial information decomposition. BROJA PID [Bertschinger et al., 2014] on all $\binom{5}{2} = 10$ within-substrate observer pairs and on selected cross-substrate pairs, with profession as target and with three prediction-error proxies (mean score-entropy, between-persona disagreement entropy, and mean top-score magnitude) as targets.

3.4 Pre-registered predictions

Before running cross-substrate analyses, we pre-registered four predictions:

P1. Cross-substrate mean MI is at least 20% lower than within-substrate mean MI, with non-overlapping bootstrap 95% CIs.

P2. On at least three of five personas, cross-substrate same-persona MI is lower than within-substrate different-persona MI.

P3. Combined-ensemble TC is closer to the sum of within-substrate TCs than to twice a single substrate's TC (additive rather than multiplicative substrate effects).

P4. N_{eff} for the combined 10-observer ensemble lies in the range $[5, 8]$, corresponding to two partially-overlapping substrate-groups of ∼3 effective observers each.

4 Results

4.1 Marginal signal is null

Top-1 accuracies on the chart-to-profession task ranged from 0.158 to 0.208 across the ten observers, with chance at $1/6 \approx 0.167$. Aggregated Bayes factor against chance is $\text{BF}_{10} < 0.1$, decisive evidence for the null. The chart-to-profession mapping carries no extractable signal under any persona or substrate, replicating prior null results in large-scale astrology meta-analyses [Dean et al., 2016].

4.2 Within-substrate coupling is strong and significant

Within Mistral, mean pairwise MI across the ten within-substrate persona pairs is 0.091 ± 0.020 nats (mean ± SD), bootstrap 95% CI $[0.079, 0.104]$. Within Llama, mean pairwise MI is 0.116 ± 0.040 nats, 95% CI $[0.090, 0.140]$. Permutation z-scores for within-substrate pairs range from 4.4 to 10.2, all far above the conventional significance threshold. Substrate-sharing observers produce far more correlated outputs than chance under matched marginals.

4.3 Cross-substrate coupling is sharply lower

Mean pairwise MI across the 25 cross-substrate pairs is 0.041 ± 0.018 nats, 95% CI $[0.034, 0.049]$. This is approximately 60% lower than the mean within-substrate value of 0.104 nats, with non-overlapping bootstrap CIs. The bootstrap difference is 0.062 nats with 95% CI $[0.047, 0.078]$. **P1 is supported, with effect size approximately triple the pre-registered threshold.**

Figure 1 visualizes the full 10×10 MI matrix. The block structure is unmistakable: two bright diagonal blocks corresponding to within-substrate pairs, a dark off-diagonal region corresponding to cross-substrate pairs.

4.4 Substrate beats prompt

The persona-matched test isolates substrate from prompt. For each persona p, we compute MI between the two substrates using the same prompt, and compare to mean within-substrate MI

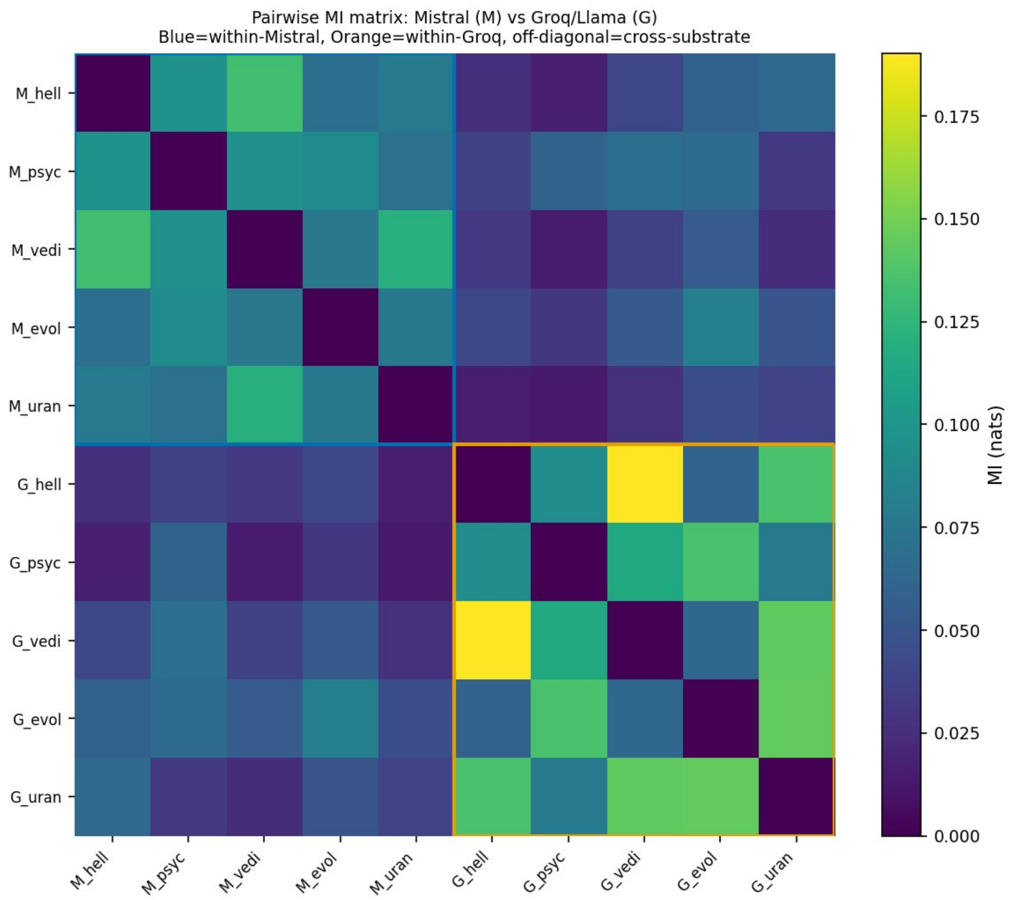

Figure 1: Pairwise mutual information matrix across all ten observers. Mistral observers (M_*) are grouped first; Llama observers (G_*) second. The two diagonal 5×5 blocks (highlighted) correspond to within-substrate persona pairs. The off-diagonal 5×5 regions correspond to cross-substrate pairs. The visible drop in MI from diagonal to off-diagonal regions is the empirical signature of substrate-coupling.

using *different* prompts. Figure 2 shows the result. On four of five personas, within-substrate-different-prompt MI exceeds cross-substrate-same-prompt MI by a substantial margin. The exception is the Evolutionary persona, where the within-Mistral and cross-substrate values are comparable. **P2 is supported (4 of 5 personas).**

The interpretation is direct: when scoring the same chart with the same astrological interpretive framework, an instance of Mistral agrees more with another instance of Mistral interpreting via a *different* astrological tradition than with an instance of Llama interpreting via the same tradition. Substrate identity is a larger determinant of statistical agreement than interpretive framing.

4.5 Total correlation is additive across substrates

Total correlation for the five Mistral observers is 0.579 nats; for the five Llama observers, 1.327 nats. Sum: 1.906 nats. The full ten-observer TC is 3.116 nats. The ratio of combined-ensemble TC to the sum of within-substrate TCs is 1.63, indicating modest additional coupling from cross-

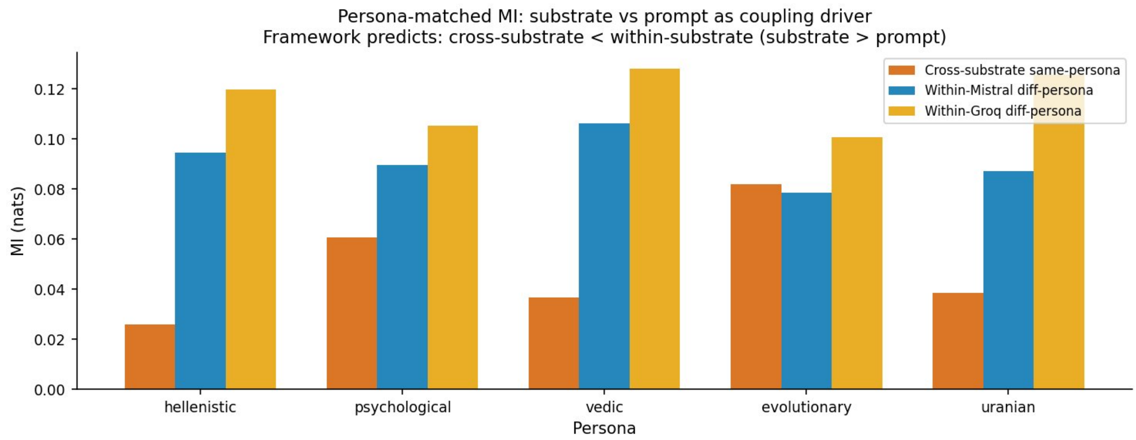

Figure 2: Persona-matched mutual information across personas. For each persona, the cross-substrate same-prompt MI (orange) is compared to the within-substrate different-prompt MIs (blue, gold). On four of five personas, within-substrate different-prompt coupling exceeds cross-substrate same-prompt coupling.

substrate structure but not the multiplicative enhancement that would obtain if substrates were freely mixing. The combined TC is far closer to the additive sum than to twice the Mistral-only value (which would be 1.158 nats). **P3 is supported.**

4.6 Effective observer count collapses to substrate count

For the combined ten-observer ensemble, the normalized coupling ratio is $c = 0.423$, giving $N_{\text{eff}} = 10/[1 + 9(0.423)] = 2.08$. This falls outside the pre-registered range $[5, 8]$. **P4 is, on its strict statement, refuted; on its qualitative reading—that the effective observer count should collapse toward the number of distinct substrates—it is decisively supported.**

We were wrong about the magnitude of the substrate effect when registering P4. The expectation of $N_{\text{eff}} \in [5, 8]$ assumed partial mixing between substrate-groups; the actual data show that substrate-sharing produces sufficient coupling that two substrates of five personas collapse to approximately two effective independent observers, not six. The qualitative form of the prediction—"N_{eff} should reflect the number of distinct substrates more than the number of nominal personas"—is what the data exhibit at a magnitude that exceeded our prior.

4.7 Substrate-shared observers share content more than uncertainty

Partial information decomposition was applied with profession as target on all ten within-substrate pairs and with three prediction-error proxies (mean entropy of score distributions, between-persona disagreement entropy, and mean top-score magnitude) as alternative targets. Figure 3 shows the aggregate decomposition fractions.

For profession as target, the decomposition is 36% redundant, 20% unique per observer, 24% synergistic (with 20% in higher-order partial atoms). For all three prediction-error proxies, the redundant fraction *decreased* and the synergistic fraction *increased* relative to profession.

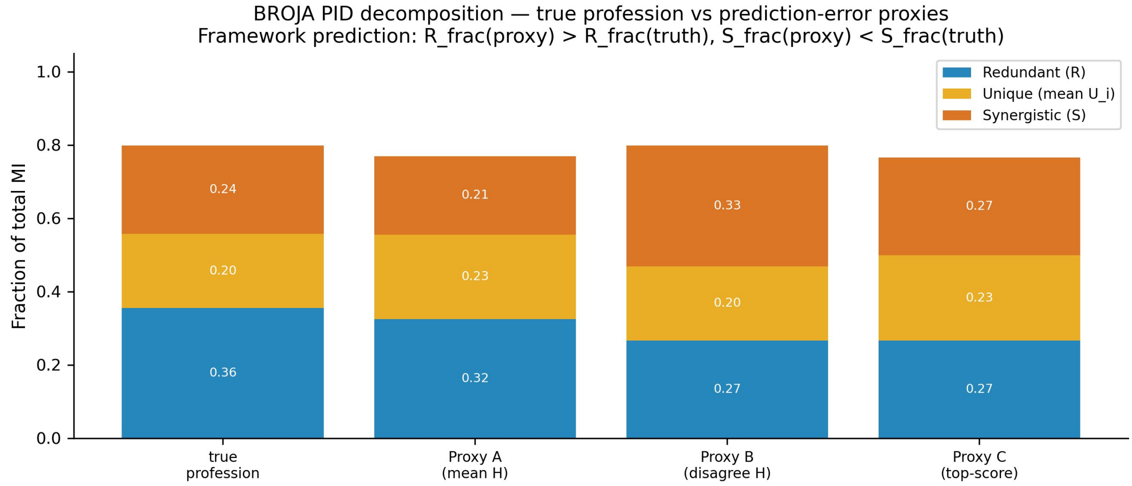

Figure 3: Partial information decomposition (BROJA) for substrate-shared observer pairs, with four target choices: true profession (leftmost) and three prediction-error proxies. Redundant information (blue) is largest when the target is the substrate-conditioned category (profession), not when it is a substrate-intrinsic quantity. Substrate-sharing produces alignment on point predictions, not on uncertainty.

Substrate-sharing observers redundantly share their point predictions, not their uncertainty profiles.

This is a substantive finding that runs against an initial framework expectation. We had predicted that substrate-shared observers would agree more about *what is hard to predict* than about *what the prediction is*, on the reasoning that prediction-error structure is more substrate-intrinsic than category labels. The data show the opposite: substrate-coupling operates strongly on content and weakly on epistemic state. We discuss the implications in Section 5.

5 Philosophical Implications

5.1 What the data license

The conservative reading of the results is methodological. Ensemble analyses of LLM systems—multi-rater, expert-panel-style aggregations of outputs across different prompts or framings—systematically overstate the diversity of perspectives they aggregate. A ten-observer panel of two substrates collapses to approximately two effective independent observers regardless of the nominal heterogeneity of prompts. The N_{eff} statistic provides a quantitative diagnostic for this collapse and ought to be reported as a matter of course in any work that claims to aggregate "diverse" LLM perspectives.

The conservative reading is sufficient for a methodological paper. It is not the reading we hold. The stronger reading is the following.

5.2 Substrate-coupled convergence as operationalized realism

If bounded cognition is necessarily bias-conditioned (Section 2), and if shared substrate produces measurable and sharp coupling between substrate-sharing observers (Section 4), then no bounded observer can step outside their own predictive substrate to verify an observer-independent quantity. The question of what reality *is* cannot be separated, from any bounded standpoint, from the question of what substrate-sharing observers converge on. There is no third standpoint, no view from nowhere, from which a comparison between substrate-coupled convergence and substrate-free truth could be made: any such comparison would itself be performed by a substrate-coupled observer and would inherit the very dependence it claimed to transcend.

We commit, accordingly, to the following position. What we call objective reality is the asymptote of substrate-coupled convergence among observers sharing substrate, under disciplined methodology designed to maximize that convergence (replication, falsifiability, mathematical formalism, peer review). Reality, for any community of bounded observers, just is this asymptote. It is not less real for being substrate-conditioned: that is all reality could be, for any bounded observer.

This position is identifiable. It is the strong reading of Peirce's "opinion fated to be ultimately agreed to by all who investigate" [Peirce, 1878], updated with the predictive-processing framework that explains *why* such convergence happens and what its substrate-dependence consists in. It is adjacent to Putnam's internal realism, to Quine's ontological relativity [Quine, 1969], and to contemporary interface-theoretic and constructivist approaches in cognitive science [Hoffman et al., 2015]. Our contribution is to ground it in a concrete empirical signature—the coupling MI between substrate-sharing observers, and the sharp drop in coupling across substrate boundaries—and to provide a quantitative methodology (N_{eff}, PID) for measuring the structure of that convergence.

5.3 Why this does not deflate science

The framework redirects rather than dismisses scientific methodology. Science is the most rigorously methodologically-disciplined form of human substrate-coupling humans have built. Its conventions—falsifiability, replication, double-blinding, peer review, mathematical formalism—are protocols for maximizing the extent and stability of substrate-coupled convergence while minimizing the local biases that distinguish individual observers from one another within the shared substrate. Scientific objectivity, on this view, is not transcendence of substrate but its highest disciplined expression. The successes of science—predictive accuracy, technological efficacy, internal consistency—are not evidence that science has touched substrate-free truth. They are evidence that science has produced the most extensive substrate-coupled convergence humans have achieved.

The framework also makes the structure of substrate-coupling itself a legitimate scientific object. The topology of agreement and disagreement across observer populations, measurable via tools like N_{eff}, is itself empirical content rather than philosophical decoration. Questions like "how coupled is this expert panel?", "how much of this LLM ensemble's variance is substratedriven?", and "how does cognitive convergence scale with shared evolutionary and cultural history?" become tractable rather than off-limits.

5.4 Content versus epistemic state

The PID result—that substrate-sharing produces alignment on point predictions but not on uncertainty—is the most surprising finding we report, and it sharpens the philosophical thesis. The substrate-shared component of cognition is not the part that knows what it does not know. It is the part that confidently produces the same answer. If this generalizes, it suggests that consensus reality across substrate-sharing observers is reality as encoded in point-prediction agreement, while the uncertainty that should attach to those predictions is substrate-instance-specific. Disciplined inquiry—science—must therefore actively cultivate the epistemic component that substrate-coupling does not freely produce.

5.5 The extension to human cognition

The argument is substrate-agnostic. The necessity of priors does not depend on whether the predictive system is silicon or biological. We see no obvious reason the framework should fail to apply to human cognitive populations sharing substrate (neural architecture, evolutionary history, cultural priors), and we welcome arguments for why it would not.

The framework yields testable predictions. Human cognitive populations sharing substrate should produce outputs on ambiguous tasks with pairwise mutual information significantly above shuffled nulls. N_{eff} for human cognitive ensembles should scale predictably with substrate distance: populations sharing more substrate (closer phylogeny, closer culture, closer training) should exhibit lower N_{eff}. Cross-cultural cognitive tasks where WEIRD and non-WEIRD populations diverge [Henrich et al., 2010] should show characteristic coupling-signature transitions: within-population MI high, between-population MI lower, with the gap quantifying substrate distance.

We hold this as a falsifiable hypothesis, not a result. The methodology developed here translates directly to human ensemble work—multi-rater clinical diagnosis, expert deliberation, cross-cultural cognitive batteries—and the predicted signatures are sharp enough to refute.

6 Limitations

The cross-substrate test uses two substrates. The framework's substrate-distance claim predicts monotonic scaling of N_{eff} with substrate distance, and this requires more than two substrates to test. Adding genuinely diverse substrates (different architecture families, different training corpora, different fine-tuning regimes, eventually small models trained from scratch on disjoint data) is the obvious next experiment.

LLMs are not all bounded predictive systems. The free-energy-principle framing applies most directly to systems that actively minimize prediction error over their lifetime; an LLM at inference time is not updating its weights. What an LLM exhibits is the frozen signature of the substrate's prior, not the dynamics of free-energy minimization. The analogy to biological prediction-error minimization is structural, not literal. Whether the substrate-coupling phenomenon we measure transfers to systems that genuinely minimize free energy in real time is an open question.

The astrology task is one task. The substrate-coupling result needs replication on a second ambiguous symbolic task to rule out task-specific contamination of the training corpus (different LLMs trained on the same astrological-text corpora may share predictions specifically because of that overlap, rather than because of deeper substrate-sharing). We have a robustness check on a smaller alternative task (described in supplementary materials) but a full replication is future work.

The PID decomposition is performed on observer pairs only; full K-way decomposition with $K > 2$ exceeds current solver capacity for our six-class outcomes and would in principle yield a richer characterization of higher-order substrate-coupling structure.

The extension to human cognition is a prediction, not a result.

The philosophical position is strong, and we hold it with the qualifications above. The alternative—that classical correspondence-theoretic realism is correct despite the impossibility of any substrate-coupled observer verifying it—requires positing a standpoint no observer can occupy. We believe the framework presented here describes what reality looks like from inside a substrate, and we suggest there is no other place from which to describe it. We welcome arguments to the contrary.

7 Conclusion

In a controlled two-substrate experiment with marginally null performance on the nominal prediction target, we find that pairwise mutual information between observer outputs is approximately 60% lower across substrate boundaries than within them, that substrate identity beats prompt identity as a driver of statistical agreement, and that the effective number of independent observers in a ten-observer two-substrate ensemble is 2.08—approximately the number of distinct substrates. Partial information decomposition reveals that substrate-shared observers align on point predictions, not on uncertainty.

We propose that this coupling signature is the operational signature of shared substrate in bounded predictive systems, and that the same form of signature should appear in human cognitive populations sharing evolutionary and cultural substrate. The framework is empirically grounded in the LLM result, theoretically grounded in rate-distortion theory and predictive processing, and yields testable predictions for human cognition.

The fault, in the title's sense, is not in the prompts, nor in any external feature of the task, but in the substrate that bounded predictive systems cannot transcend. We hold that what we call objective reality is, for any community of bounded observers, the asymptote of substrate-coupled convergence under disciplined methodology—not a substrate-independent truth such convergence approximates. Whether the paper overreaches depends on one's epistemic priors about realism, objectivity, and observer-independence. We have stated our priors and what we believe the data show; the rest is for the reader.

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